Phase 10: Sprint Metrics and Burndown Analysis

Course Code: 20CYS401 | **Project**: Online Book Review and Recommendation Platform

---

10.1 Sprint Summary & Velocity

Metric	Sprint 1 (Foundations & Core Features)	Sprint 2 (Access Control & Hardening)	Total Project
Planned Story Points	18	26	44
Completed Story Points	18	26	44
Commit Count	Initial Scaffold + App Implementation	IDOR Refactoring + Security Hardening	4 commits
Velocity	18 pts/sprint	26 pts/sprint	22 pts avg
Defect Density	0 critical leaks to main	0 critical leaks to main	0 escaped defects

---

10.2 Sprint Burndown Data

Sprint 1 Burndown (18 Points Total)

- **Day 1**: 18 pts remaining (Sprint Planning, Architecture design)
- **Day 2**: 13 pts remaining (User registration and authentication completed - BRP-1, 5 pts)
- **Day 3**: 8 pts remaining (Book search implemented - BRP-2, 5 pts)
- **Day 4**: 5 pts remaining (Book details & ratings display completed - BRP-3, 3 pts)
- **Day 5**: 0 pts remaining (Review submission completed - BRP-4, 5 pts)

Sprint 2 Burndown (26 Points Total)

- **Day 1**: 26 pts remaining (Sprint 2 Planning, IDOR threat modeling)
- **Day 2**: 18 pts remaining (Edit/delete ownership enforcement & IDOR test - BRP-5, 8 pts)
- **Day 3**: 13 pts remaining (Reading list private shelf - BRP-6, 5 pts)
- **Day 4**: 8 pts remaining (Admin catalog CRUD - BRP-7, 5 pts)
- **Day 5**: 3 pts remaining (Review moderation & Audit logging - BRP-8 & BRP-9, 10 pts)
- **Day 6**: 0 pts remaining (Containerization, probes, security scans - BRP-10, 3 pts)

10.3 Burndown Chart (Mermaid Representation)

```
xychart-beta
title "Sprint 2 Burndown Chart (Story Points vs Days)"
x-axis ["Day 1", "Day 2", "Day 3", "Day 4", "Day 5", "Day 6"]
y-axis "Story Points Remaining" 0 --> 30
line [26, 18, 13, 8, 3, 0]
```

10.4 Sprint Retrospectives & Continuous Improvement

Sprint 1 Retrospective

- **What went well**: SQLAlchemy ORM prevented SQL injection completely; unit tests provided high confidence for password hashing.

- **What went wrong**: Database relative path issues initially caused SQLite file locking during test executions; resolved by switching to absolute paths in `Config`.

- **Action item**: Enforce object-level permission tests for all upcoming mutable endpoints in Sprint 2.

Sprint 2 Retrospective

- **What went well**: IDOR vulnerability was explicitly modeled, reproduced in an isolated demonstration branch, and proven mitigated on `main`.

- **What went wrong**: Bandit scan flagged test configuration `SECRET_KEY`; identified as Low severity test-only artifact and documented in security audit.

- **Action item**: Complete comprehensive traceability matrix across all phases.